

Day - 16

SETS

in Python



 **SKILL COURSE**



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What is a Set?

- A **set** is like a basket where items cannot repeat. If you put duplicate items, the basket will **keep only unique items**.
- Example:
"Apple, Apple, Banana" → becomes → "Apple, Banana"
- ✓ **Technical Definition**
- A set is an **unordered, mutable** collection that stores **unique values only**.
Created using { }.
- Example: {"apple", "banana", "mango"}

Why Sets Are Important for Data Analysts?

- Remove duplicates from data
- Extract unique values from columns
- Compare two lists
- Find items missing from one dataset
- Improve cleaning speed
- Faster membership checking
- Useful in data validation
- Useful in merging and intersection tasks

Set

- Creating a Set
 - `fruits = {"Apple", "Banana", "Apple", "Mango"}`
 - `print(fruits)`
- Output:
 - `{'Apple', 'Banana', 'Mango'}`
- Adding Items to a Set
 - `fruits.add("Orange")`

Sets

- Removing Items
 - `fruits.remove("Banana")`
 - `fruits.discard("Papaya")` # No error if not exists
- Set Does NOT Allow Indexing
 - `# fruits[0]` ❌ Error
 - Sets are unordered → cannot access by index.

Set Operations

Union (combine all unique items)

`a = {1, 2, 3}`

`b = {3, 4, 5}`

`print(a | b) # {1,2,3,4,5}`

Intersection (common items)

`print(a & b) # {3}`

Difference

`print(a - b) # {1,2}`

Symmetric Difference

`print(a ^ b) # {1,2,4,5}`

Sets

Looping Through a Set

for item in fruits:

print(item)

Remove Duplicate City Names

```
cities = ["Mumbai", "Pune", "Mumbai", "Delhi",  
"Pune"]
```

```
unique = set(cities)
```

```
print(unique)
```

Real Data Analyst Use Cases

Find Missing Values Between Two Lists

```
list1 = {"Laptop", "Mouse", "Keyboard"}
```

```
list2 = {"Laptop", "Keyboard"}
```

```
missing = list1 - list2
```

```
print(missing)
```


Find Common Items Between Two Departments

```
deptA = {"SQL", "Excel", "Python"}  
deptB = {"Excel", "Power BI", "Python"}  
  
common = deptA & deptB  
print(common)
```

Sets Practice

Extract Unique Product Categories

```
categories = ["Electronics", "Grocery",  
             "Electronics", "Fashion"]  
unique_cats = set(categories)  
print(unique_cats)
```

Convert Set Back to List

```
unique_list = list(unique_cats)
```

Assignments for Students

Basic

Create a set of 5 colors.

Add and remove items.

✓ Intermediate

Remove duplicate countries from a list.

Find common subjects between two students.

✓ Advanced

Find missing values between two product lists.

Write a program that extracts unique words from a sentence.

